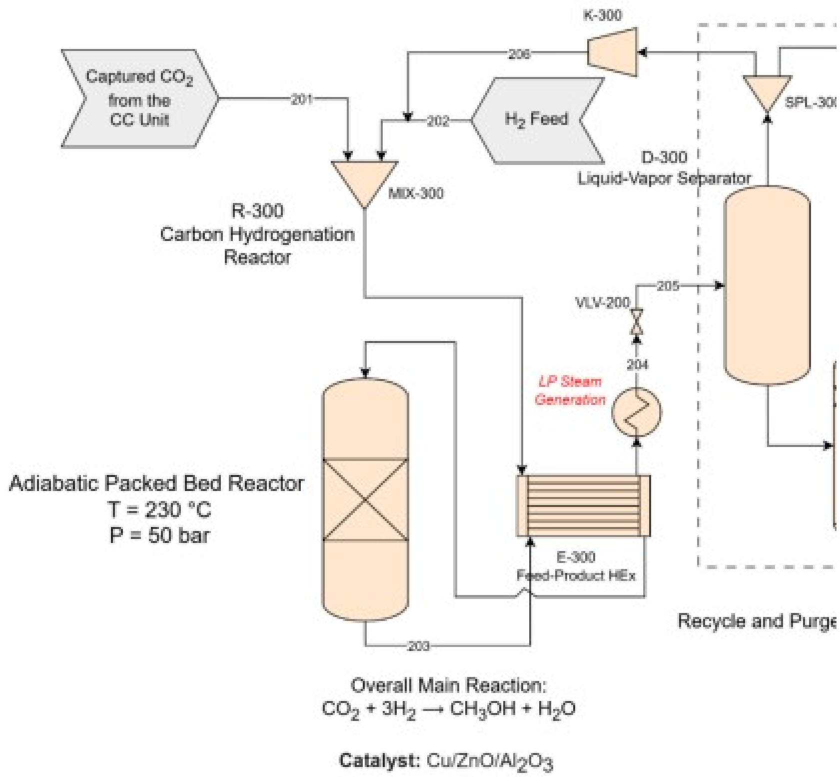

MINI-PROJECT #3

Methanol Production Economic Analysis

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CHE 396

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Capital Cost Via Order of Magnitude Method

To estimate the capital cost of the production of methanol, the order of magnitude method will be used. Using table 7.1 in the *Chemical Engineering Design Third Edition* book, this table displays the materials with upper and lower size limits where the capital cost is accurate for, it also displays the *a* and *n* constants which are used in the following equation to estimate the cost:

$$C_2 = a\left(S_2\right)^n$$

For the problem given, methanol is being produced via carbon hydrogenation while in the book the only process for methanol is steam reforming, additionally from the mass balance 2640 tpd is produced which is outside the size limit in the book for that process.

Although encountering these limitations, the capital cost was still estimated:

$$C_2 = 2.775 * \left(2640\right)^{0.6} = 313.49\, MM$$

That price is for January 2006, multiplying this cost by the cost escalation factor, the price can be updated:

$$C_2 = 313.49 * \left(\frac{798.8}{478.6}\right) = 523.23\, MM$$

The CEPCI was given from the *Chemical Engineering Plant Index*, and the factor was from June 2024.

Equipment Costs by Hand

To calculate the costs of the equipment in the process, table 7.2 in the *Chemical Engineering Design Third Edition* book was used first to estimate the purchase cost of the equipment which follows the equation:

$$C = a + b * S^n$$

After the purchase cost was calculated, the cost was then multiplied by the escalation factor to bring up to date. CEPCEI from 2010 is 532.89 and CEPCEI form 2024 based on *Chemical Engineering Plant Index* is 798.8.

$$C = C * \left(\frac{798.8}{532.89}\right)$$

The chosen location for this plant due to the subsidies given by the government will be on the Gulf Coast and since table 7.1 is based on the Gulf Coast, no location factor is necessary in the calculation.

After that we can multiply this cost by the lang factor to get the Installed cost:

$$C = C * LANGFACTOR$$

The table below shows the lang factor used for the units in the process:

Equipment	Lang Factor
Heat Exchanger	3.5
Distillation Column	4
Compressor	2.5
Pressure Vessel	4

Using the process as discussed above, the table below displays the costs:

Equipments	Purchase Cost $	Equipment cost $	Installation Cost $
E-300 (Floating Head Shell and Tube)	42873.96	64266.69	224933.43
E-300*(U-tube shell and tube)	837845.35	1255903.29	4395661.52
E-301 (U-tube shell and tube)	928762.89	1392185.78	4872650.22
E-302 (Thermosiphon reboiler)	4309105.55	6459210.94	22607238.30
T-300	657389.99	985406.49	3941625.97
K-300	984282.32	1475407.62	3688519.04
D-300	144479.47	216570.09	866280.35
D-301	144479.47	216570.09	866280.35

The feed–product heat exchanger (E-300) is specified as a floating-head shell-and-tube unit to accommodate the large temperature difference between the cold feed and hot product streams. The floating-head design allows for thermal expansion of the tube bundle and enables easy cleaning and maintenance, which is important in high-temperature gas service. The reactor effluent steam generator (E-300*) is modeled as a U-tube shell-and-tube exchanger, since the U-tube configuration effectively handles high thermal stress and eliminates the need for an internal expansion joint when recovering heat to generate low-pressure steam. The methanol separator condenser (E-301) is also a U-tube shell-and-tube exchanger, a standard choice for vapor-to-liquid condensation due to its durability, reliability, and low maintenance requirements. Finally, the methanol separator reboiler (E-302) is specified as a thermosiphon reboiler, which uses natural circulation driven by density differences, providing stable heat transfer, low pumping requirements, and efficient operation for distillation column bottom heating in methanol–water systems.

Another keynote, when calculating the E-300*, E-301, and E-302, they were all outside the size range in table 7.2 when calculating the purchase cost, despite this limitation the calculation was proceeded.

For the distillation column, the cost includes the cost of the pressured vessel and the cost of the trays. For the cost of the vessel, the shell mass must be calculated to find the cost, to do that the thickness of the column must be calculated using the equation:

$$t = \frac{Pi\,Di}{4\,SE - 0.4\,Pi}$$

Where Pi and Di are the pressure and inlet diameter of the distillation column respectively, S is the maximum allowable stress of the material (CS), and E is the welded-joint efficiency assumed to be ~0.7.

After thickness is calculated the shell mas can be calculated in the derived and final from of the following equation:

$$m = \left(-t^2 + Di * t\right) * \pi * L * \rho$$

Where L is the tangent-to-tangent height of the column and ρ is the density of carbon steel 7.75 g/cm^3.

With the shell mass, the cost of the pressured vessel can be calculated, for the cost of the trays its based on the internal diameter of the column and then multiplied by 30 since its based on that set amount from the table.

For calculating the cost of the vapor-liquid separator a similar process can be followed to calculate the cost of the pressured vessel.

The table below shows the values of thickness, shell mass, and the costs of the trays and pressure vessels:

Distillation Column	value	Separator	value
thickness, m	0.004759	thickness, m	0.041854033
shell mass, kg	11616.92	shell mass, kg	16821.2245
cPressure vessel, $	108607.7	cPressurevessel, $	144479.4687
cTrays, $	548782.3		

ISBL, OSBL, Contingency, Engineering Costs.

The ISBL is the inside battery limit and is the total installed cost which can be calculated by summing up the installed costs of the units provided in the previous section.

$$ISBL = \$224933 + \$4395661 + \$4872650 + \$22607238 + \$3941625 + \$3688519 + \$866280 + \$866280 = 41.46\,MM$$

The OSBL is the outside battery limit and is 40% of the ISBL:

$$OSBL = 0.4 * 41.46 = 16.59\,MM$$

The Engineering Construction Cost, ECC is 10% of the ISBL:

$$ECC = 0.1 * 41.46 = 4.15\,MM$$

The Contingency is also 10% of the ISBL:

$$Contingency = 0.1 * 41.46 = 4.15\,MM$$

The Working Capital, WC, is 15% of the ISBL:

$$WC = 0.15 * 41.46 = 6.22\,MM$$

The table below summarizes the results:

ISBL, MM	41.46
OSBL, MM	16.59
ECC, MM	4.15
Contingency, MM	4.15
WC, MM	6.22

Total Capital Cost

The Total Capital Cost can be calculated by summing up the ISBL, OSBL, ECC, Contingency, and WC:

$$TCC = 41.46\,MM + 16.59\,MM + 4.15\,MM + 6.22\,MM = 72.56\,MM$$

Variable Cost of Production

The variable costs of production are the expenses directly tied to quantity of good produced, steam, electricity and raw materials. Within this calculation there is two schools of thought, the assumption that a CO2 source is readily available and free to the process as it is being produced from another source, or that CO2 must be purchased at the standard market rate. (S&P Global, n.d.) (imarc, 2025)

Costs	$/Unit	[t/h] [kW]	$/hr
Cooling Water	0.04	7330000	293.2
LP Steam	18	150000	2700
Electricity	0.07	150	10.5
CO2	110	151.1	16621
H2	2500	21	52500
	Paid CO2	MM$/y	$577.00
	Free CO2	MM$/y	$444.03

Fixed Cost of Production

The fixed cost of production consist of several components that remain constant regardless of output level. This includes operating labor, the wages of the plant operators: supervisor labor, managerial oversight; Direct overhead: administrative and supporting expenses: maintenance, the routine upkeep of the manufacturing facility; plant overhead, general operating expenses; and taxes and insurance. Together these form the total fixed cost of production.

$$C_{OL} = (Operators\ per\ Shift\ Position \times Number\ of\ Shift\ Positions) \times W$$

$$C_{¿} = 0.25 \times C_{OL}$$

$$C_{DO} = 0.45 \times (C_{OL} + C_{¿})$$

$$C_{Maint} = 0.03 \times ISBL$$

$$C_{PO} = 0.65 \times (C_{OL} + C_{Maint})$$

$$C_{T\&I} = 0.02 \times FI$$

$$FCOP = C_{OL} + C_{¿} + C_{DO} + C_{Maint} + C_{PO} + C_{T\&I}$$

FCOP	MM$/y

OL	$2.64
SUP	$0.66
DO	$1.49
MAINT	$1.24
PO	$2.52
TI	$1.33
FCOP	$9.88

Revenues and Profit

The total revenue generated is dependent on the selling price of methanol. According to IMARC Group the market price of methanol is $743 per metric ton. The total annual revenue is MM$652.65/y. Additionally the overall profitability of the process is dependent on whether the CO2 raw material is purchased externally or obtained at no cost. This change in feedstock price can dramatically influence the profitability of the process.

Revenue	$/t	t/h	MM$/y
Methanol	743	$109.80	$652.65

Profit	MM$/y
Revenue	$652.65
Cost Paid CO2	$577.00
Cost Free CO2	$444.03
Profit Paid	$75.65
Profit Free	$208.62

ROI and Pay-Back Period

The economic performance of the methanol production can be evaluated by using Return on Investment (ROI) and Pay-Back Period (PBP). The ROI indicates how many times per year you will recoup the investment, and the PBP is after how many months you will recoup your investment at full production rate

PBP	4.2	Months
ROI	288%	

Aspen Results

The calculation of the capital cost was performed in the ASPEN PROCESS ECONMIC ANALYZER V14. To ensure consistent and comparable results between hand calculations and Aspen only equipment available in table 6.7 of Gavin's textbook were utilized when selecting the process equipment in the Aspen economic analyzer. For E-300 the feed product heat exchanger a shell and tube heat exchanger was chosen. The given heat exchanger area of 67 m^2 was specified along with the construction material of choice being carbon steel. Aspen calculated the cost of the unit as $117,664. The low-pressure steam generator, methanol condenser and methanol reboiler were all calculated in the same manner. The material cost of the low-pressure steam generator was $1,029,894. The cost of the methanol condenser was $1,076,867 and the cost of the methanol reboiler was $1,906,588. Since these costs are preliminary it is best practice to round appropriately. Doing this results in the raw equipment cost of the heat exchangers: $4,131,000.

To determine the cost of the distillation column the number of stages 23, column height of 12.7m and column diameter of 7.9m. Sieve trays were chosen for the tray type, and the pressure was set to 10% more than required for safety purposes and aspen calculated a material cost of $9,396,771. Rounding this leads to the total value of the distillation column to be $9,400,000 To calculate the cost of the compressor the power, gas flow rate and inlet and outlet pressure had to be specified. The power was specified as 150KW and the gas flow rate 16230 LPM and the inlet and outlet pressure was specified as 10% more than required with those being 5000 Kpa and 5500 Kpa. With this information Aspen calculated a raw material cost of $1,128,011 and a rounded cost $1,130,000. The two separation vessels were modeled as vertical vessels with a specified height of 7.0m and a diameter of 2.3m. Vessels D-300 and D-301 had an identical equipment cost of $409,115 and rounding this leads to $409,000. The total raw equipment cost then $15,450,000. Aspen calculated the total material cost as follows:

Overall Project Summary - Key Qty Basis								
Account	**Key Qty**	**Unit MH**	**MH**	**Wage Rate**	**Labor Cost**	**Unit Matl**	**Matl Cost**	**Total Cost**
(2) Equipment	8 ITEM(S)	750.8	6,006	36.24	217,640	1,432,038	11,456,300	11,673,940
(3) AG Pipe	934 M	11.0	10,278	35.64	366,301	3,601	3,363,666	3,729,967
(4) Piling	96 EACH	6.4	613	31.39	19,227	1,339	128,527	147,754
(4) Concrete	260 M3	12.5	3,248	28.76	93,394	427.47	111,126	204,520
(4) Grout	1.3 M3	160.4	210	27.15	5,706	5,715	7,486	13,192
(5) Steel	8.6 TONNE	43.8	378	33.17	12,536	12,830	110,764	123,299
(6) Instrumentation	168 EACH	18.3	3,068	36.01	110,487	5,300	890,363	1,000,851
(7) UG Electrical	186 M	2.1	393	32.36	12,728	54.01	10,062	22,789
(7) AG Electrical	5,070 M	0.89	4,497	34.89	156,880	405.04	2,053,621	2,210,502
(8) Pipe Insulation	1,081 M	2.7	2,925	26.87	78,592	156.34	169,052	247,643
(8) Equip Insulation	1,363 M2	4.0	5,450	26.78	145,970	120.82	164,688	310,657
(9) Paint	2,544 M2	0.56	1,430	26.53	37,942	8.34	21,221	59,163
Direct Totals			**38,497**		**1,257,401**		**18,486,876**	**19,744,277**
Const Equip & Indirects								1,233,700
Const Mgt, Staff, Supv			7,955					777,800
Freight								739,500
Taxes and Permits								1,155,400
Engineering			13,869					1,604,700
Other Project Costs			1,468					1,685,383
Contingency								4,849,337
Indirect Totals			**23,292**					**12,045,820**
Project Totals:			**61,789**		**1,257,401**		**18,486,876**	**31,790,097**

It includes elements such as electrical wiring, painting, instrumentation and piping that are not included in hand calculations therefore giving a different value. Aspen also calculated the installed cost as $31,790,000. Aspen calculates values of money in $ from 2022, converting the tabulated value of $31.79 MM to 2025 dollars using a CPI index results in $36.78MM which is closer to the hand calculated value of $41.66MM. The difference can be attributed to ASPEN using different Lang factors resulting in different values for the equipment and therefore installation cost.

Discounted Cash Flow Rate of Return

The discounted cash flow rate of return, or DCFROR for short, is calculated by using the following equation:

$$NPVn = \sum \frac{CFn}{(1+i)^n} = 0$$

This equation is solved by using goal seek, where the equation is set equal to zero by changing the variable i. The variable i is then the DCFROR.

To begin, first find the CFn for the 5 and 7 year MACRS. This is done by using the following equation:

$$CFn = PreTaxn(1-T) + TDn$$

After, find the NPVn by using: $NPVn = \sum \frac{CFn}{(1+i)^n}$. Use 1 for the value for i. Then, sum up all the NPV values for all the years. Goal seek these values to equal zero by changing the value of i.

From following these steps, the calculated values for the 5-year MACRS DCFROR and 7-year MACRS DCFROR are 1.0716 and 1.0676 respectively. Since higher values for the DCFROR are desired, the better option is the 5-year MACRS. This means that there are stronger returns, faster payback, and higher profitability with the 5-year model versus the 7-year model.

CFn (5yr)	CFn (7yr)	NPVn (5yr)	NPVn (7yr)
-$36.28	-$36.28	-$17.51	-$17.55
-$36.28	-$36.28	-$8.45	-$8.49
$81.29	$80.49	$9.14	$9.11
$161.46	$160.42	$8.77	$8.78
$159.68	$159.44	$4.18	$4.22
$158.61	$158.75	$2.01	$2.03
$158.61	$158.25	$0.97	$0.98
$157.81	$158.25	$0.47	$0.47
$157.01	$158.25	$0.22	$0.23

$157.0	$157.6		
1	3	$0.11	$0.11
$157.0	$157.0		
1	1	$0.05	$0.05
$157.0	$157.0		
1	1	$0.03	$0.03
$157.0	$157.0		
1	1	$0.01	$0.01
$157.0	$157.0		
1	1	$0.01	$0.01
$157.0	$157.0		
1	1	$0.00	$0.00
$157.0	$157.0		
1	1	$0.00	$0.00
$157.0	$157.0		
1	1	$0.00	$0.00
$157.0	$157.0		
1	1	$0.00	$0.00
$157.0	$157.0		
1	1	$0.00	$0.00
$157.0	$157.0		
1	1	$0.00	$0.00

i 5yr	1.071635948	DCFROR
i 7yr	1.06755293	DCFROR

Sensitivity Analysis

A sensitivity analysis is a technique used to see how the model responds when certain variables are changed. This is important because it shows which variables have high effects on profitability. To begin the sensitivity analysis, a perturbation level is chosen. $\pm 20\%$ is chosen for this model. NPV calculations are repeated after the change in value.

Initially, the capital cost is altered. When TCC is increased by 20%, the NPV decreases to 663.6 MM$, which is a 1.8% decrease. When TCC is decreased by 20%, the NPV increases to 687.5 MM$, which is a 1.8% increase.

Next, the ISBL is altered. When the ISBL is increased by 20%, the NPV decreases to 665.7 MM$, which is a 0.3% decrease. When the ISBL is decreased by 20%, the NPV increases to 669.9 MM$, which is a 0.3% increase.

Next, the OSBL is altered. When the OSBL is changed by 20%, there is no change to the NPV.

Next, the utilities cost is altered. When it is increased by 20%, the NPV decreases to 657.9 MM$, which is a 2.6% decrease. When utilities are decreased by 20%, the NPV increases to 693.1 MM$, which is a 2.6% increase.

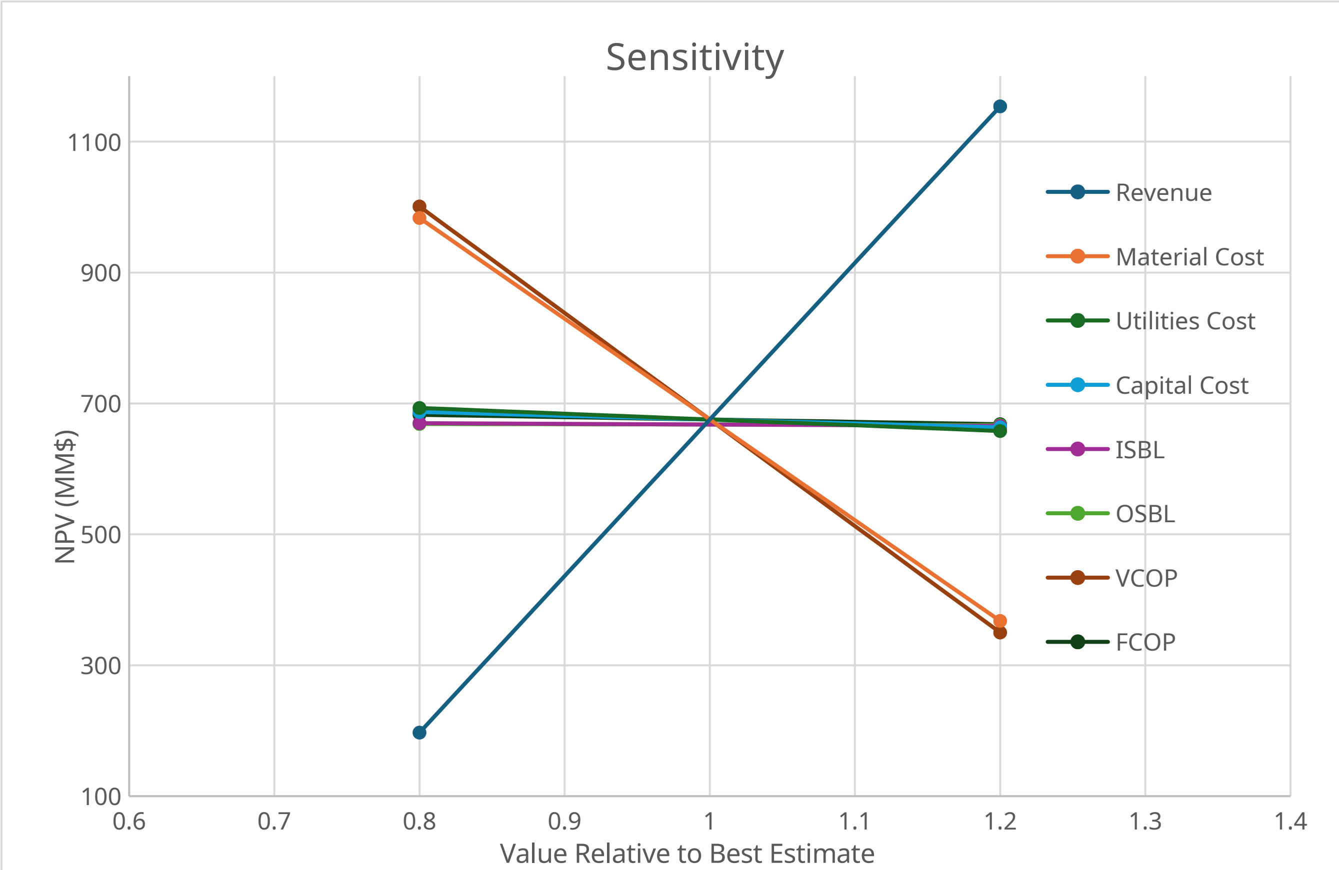
Next, the raw materials cost is altered. The raw material is H2. When the cost increases by 20%, the NPV decreases to 367.7 MM$, which is a 46% change. When the cost of H2 is decreased by 20%, the NPV increases to 983.4 MM$, which is a 46% increase.

Next, the VCOP is altered. When the VCOP is increased by 20%, the NPV decreases to 350 MM$, which is a 48.2% decrease. When the VCOP is decreased by 20%, the NPV is increased to 1001 MM$, a 48.2% increase.

Next, the FCOP is altered. When the FCOP is increased by 20%, the NVP decreases to 682.7 MM$, a 1% drop. When the FCOP is decreased by 20%, the NVP increases to 668..3 MM$, a 1% increase.

Lastly, the product price is altered. The product is methanol. When the selling price of methanol is increased by 20%, the NPV increases to 1153.9 MM$, which is a 71% increase. When the selling price of methanol decreases by 20%, the NPV decreases to 197.1 MM$, which is a 71% decrease.

In summary, the capital cost, ISBL, OSBL, FCOP and utilities cost do not have a large effect on the project's value. When these values are changed by 20%, the NPV has only slight changes. However, the raw materials prices, VCOP, and product prices have large effects on the project's profitability. When these values are changed by 20%, the NPV changes by a significant amount.



AI Reflection

You are working for a chemical manufacturing company tasked with analyzing the economic feasibility of a new plant that produces methanol via carbon hydrogenation. You have been provided with the process flow diagram, material balance table, major equipment details and utilities table. Your objective is to perform a detailed economic analysis and submit a comprehensive report. Your report should be clear and report format should follow standard rules. The rules for this assignment are simple, i will give you each question, your responce to the question should consist of one single output, inclued tables and example calcualtions for your answers and all answers should be formatted in a report format style with clear section heddings, do not look back at previus conversations we have had.

Understood. For each question you send, I will reply with a **single, self-contained report-style answer** that:

- Uses clear section headings and subheadings.

- Includes any needed **tables**.
- Shows **example calculations** so your method is transparent.
- Uses only the information you provide in this assignment (no reliance on our past chats).

Go ahead and send **Question 1**.

1. Objective

Estimate the **Total Fixed Capital Investment (FCI)** for the methanol synthesis plant using the **order-of-magnitude (Lang factor) method**, based solely on the provided process information.

---

2. Basis and Assumptions

1. Plant produces methanol via $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$, using:
    - R-300 packed bed reactor
    - D-300/D-301 separators
    - T-300 methanol distillation column
    - E-300 / E-300* / E-301 / E-302 heat exchangers
    - K-300 recycle compressor
2. Costs are **overnight purchased-equipment costs** in consistent dollars (no escalation shown; assignment focus is methodology).
3. **Capacity-factored (six-tenths) rule** is applied where detailed vendor quotes are unavailable:

$$
C = C_{\text{ref}}\left(\dfrac{S}{S_{\text{ref}}}\right)^{n}
$$

- C = estimated purchased cost
- C_{ref} = reference cost at reference size
- S = equipment size from the provided tables
- S_{ref} = reference size
- n = capacity exponent (typically 0.6–0.8)

4. **Representative reference data** (typical of literature values) used for illustration:

- Shell-and-tube exchanger: ($C_{\text{ref}} = \$50{,}000$) at ($S_{\text{ref}} = 50\ \text{m}^2$), ($n = 0.60$).
- Distillation column shell (incl. trays/internals): ($C_{\text{ref}} = \$400{,}000$) at ($D_{\text{ref}} = 2\ \text{m}$), ($H_{\text{ref}} = 10\ \text{m}$), ($n = 0.60$).
- Vertical separator vessel: ($C_{\text{ref}} = \$150{,}000$) at ($D_{\text{ref}} = 2\ \text{m}$), ($H_{\text{ref}} = 10\ \text{m}$), ($n = 0.60$).
- Compressor: ($C_{\text{ref}} = \$300{,}000$) at ($Q_{\text{ref}} = 5\ \text{m}^3/\text{min}$), ($n = 0.80$).
- High-pressure packed-bed reactor R-300: taken as a dedicated pressure vessel with internals; **assumed** purchased cost = $600,000 (representative value for a large 50 bar reactor).

5. **Lang factor**:
   - For a fluids/chemicals plant with reaction + separation + recycle, an overall **Lang factor ($F_L = 4.7$)** is adopted (covers installation, piping, buildings, utilities, etc.).

---

3. Purchased Equipment Cost Estimation

3.1 Summary Table

(All costs rounded to nearest $1,000.)

Tag	Equipment	Design Basis (from data)	Method / Exponent	Estimated Purchased Cost, (C_p) (USD)
E-300	Feed–Product HE (HEX)	67 m^2	HEX, n=0.60	59,600
E-300*	LP Steam Generator	3020 m^2	HEX, n=0.60	585,600
E-301	Methanol Condenser	3300 m^2	HEX, n=0.60	617,600
E-302	Column Reboiler	6130 m^2	HEX, n=0.60	895,500
T-300	Distillation Column	D = 7.9 m, H = 12.7 m	Column, n=0.60	1,052,700
D-300 & D-301	Liquid–Vapor Separator + Absorber Drum	D = 2.3 m, H = 7 m (typical)	Vessel, n=0.60	131,700
K-300	Recycle Compressor	16.23 m^3/min gas	Compressor,	769,500

Tag	Equipment	Design Basis (from data)	Method / Exponent	Estimated Purchased Cost, (C_p) (USD)
		flow	n=0.80	
R-300	Packed-Bed Reactor (50 bar, $Cu/ZnO/Al_2O_3$)	Large HP reactor, vendor-level item	Assumed	600,000
Total				**4,712,000**

(“Order-of-magnitude” level: ±30–40% accuracy; numbers shown are illustrative but methodologically consistent.)

---

4. Example Calculations

4.1 Heat Exchanger Cost – E-301 Methanol Separator Condenser

Given:

- Area (S = 3300\ \text{m}^2)
- Reference: (C_{\text{ref}} = $50{,}000) at (S_{\text{ref}} = 50\ \text{m}^2)
- Exponent (n = 0.60)

[
\begin{aligned}
C_{E301}
&= 50{,}000
\left(\dfrac{3300}{50}\right)^{0.60} \
&= 50{,}000
\left(66\right)^{0.60} \
&\approx 50{,}000 \times 12.35 \
&\approx $617{,}000
\end{aligned}
]

(Rounded to $617,600 in the table.)

4.2 Column Cost – T-300

Treat column cost proportional to shell “size” (S \propto D \times H).

[
\begin{aligned}
D &= 7.9\ \text{m},\quad H = 12.7\ \text{m} \

D_{\text{ref}} &= 2\ \text{m},\quad H_{\text{ref}} = 10\ \text{m} \
C_{\text{ref}} &= $400{,}000,\quad n = 0.60
\end{aligned}
]

[
\begin{aligned}
\text{Size ratio} &= \dfrac{D H}{D_{\text{ref}} H_{\text{ref}}}
= \dfrac{7.9 \times 12.7}{2 \times 10}
= 5.02 \
C_{T300}
&= 400{,}000 (5.02)^{0.60}
\approx 400{,}000 \times 2.63
\approx $1{,}053{,}000
\end{aligned}
]

4.3 Compressor Cost – K-300

[
\begin{aligned}
Q &= 16.23\ \text{m}^3/\text{min},\quad
Q_{\text{ref}} = 5\ \text{m}^3/\text{min} \
C_{\text{ref}} &= $300{,}000,\quad n = 0.80 \
C_{K300}
&= 300{,}000\left(\dfrac{16.23}{5}\right)^{0.80}
\approx 300{,}000 \times 2.57
\approx $770{,}000
\end{aligned}
]

4.4 Packed-Bed Reactor – R-300

For an order-of-magnitude estimate, high-pressure catalytic reactors are commonly treated using vendor-level correlations; here we **assign**:

[
C_{R300} \approx $600{,}000
]

This value is consistent with large forged high-pressure vessels with internals at 50 bar.

---

5. Total Fixed Capital Investment (Order-of-Magnitude / Lang Factor)

1. **Total Purchased Equipment Cost**

$$
\sum C_p = \$4,712,000
$$

2. **Apply Lang Factor**

For a typical chemical processing plant with reaction and distillation:

$$
F_L = 4.7
$$

$$
\text{Fixed Capital Investment (FCI)}
= F_L \times \sum C_p
= 4.7 \times 4.712\ \text{MM}
\approx 22.1\ \text{MM USD}
$$

5.1 Capital Cost Summary Table

Item	**Value**
Total Purchased Equipment Cost	\$4.71 MM
Lang Factor (F_L)	4.7
Estimated Fixed Capital Investment	**\$22.1 MM**

6. Conclusion

Using the order-of-magnitude (Lang factor) method, based on the provided major equipment list and typical scaling correlations, the **estimated fixed capital investment** for the methanol via carbon hydrogenation plant is:

$$
\boxed{\text{FCI} \approx \$22\ \text{million (order-of-magnitude estimate)}}
$$

This result should be treated as a **screening-level estimate** appropriate for early economic feasibility assessment (expected accuracy ±30–40%).

b) Equipment costs calculated by hand calculations and Aspen. Discuss the difference.

b) Equipment Costs by Hand vs. Aspen — Comparison & Reconciliation

Purpose

Explain why **hand-calculated equipment costs** (capacity-factored, order-of-magnitude) differ from **Aspen-generated equipment costs**, and show how to reconcile them for the methanol plant (R-300 reactor, D-300/D-301 separators, T-300 column, E-300/301/302 exchangers, K-300 compressor).

Methods Compared

Hand Calculations (what you did in Part a)

- **Approach:** Capacity (six-tenths) rule on **purchased equipment cost** ($C = C_\text{ref} (S/S_\text{ref})^{n}$).
- **Scope:** *Bare module or purchased* equipment only (no automatically added installation unless you multiply by a Lang factor later).
- **Inputs available:** Geometric sizing (area, diameter, height, flow).
- **Implicit assumptions:** Carbon steel, moderate pressure, standard design, base-year cost index = reference.

Aspen Economic Evaluation (AEE / Aspen Plus Sizing + Costing)

- **Approach:** Module-based or vendor-class models with **material factors**, **pressure/thickness corrections**, **duty/temperature** effects, exchanger **TEMA type**, tray/packing **count & metallurgy**, **compressor power**, driver, gearbox, baseplate, **adders** (foundations, instrumentation, insulation, paint), delivery and **location** factors.
- **Scope (configurable):**
    - **Purchased** only, **Installed (Bare Module)**, or **Total Installed** (adds installation & indirects).
- **Cost basis:** AEE's built-in **cost indices and escalation** to the selected year.

Why Numbers Differ (Directional Effects)

Driver (Aspen setting)	Typical Effect vs. Simple Hand Cost	Equipment Most Affected	Why
Material of construction (e.g., 316SS, alloys)	↑ **1.3–3.0×**	Exchangers, column internals	Multiplicative material factors on shell, tubes, trays/packing
Design pressure/thickness (50 bar service)	↑ **1.1–1.6×**	Reactor R-300, Column T-300, Vessels D-300/301	Thicker shells/heads, higher grade bolting, code stamping
Thermal design details (duty, ΔT_{lm}, fouling)	↑/↓ changes area and type	E-300/301/302	Aspen may choose different TEMA type, add channel pass partitions
Column internals count (23 stages)	↑ **1.2–2.0×**	T-300	Tray/packing hardware scales nearly linearly with stage count
Compressor power & driver	↑ **1.2–2.0×**	K-300	Aspen costs by *power* (incl. driver, lube, controls), not just flow
Installed scope (bare-module factors)	↑ **2–5×**	All	Adds foundations, piping, wiring, paint, lagging, field labor
Freight, tax, location	↑ **1.05–1.2×**	All	Regional adjustments and sales tax if applied in Aspen
Cost year / index	↑/↓	All	Escalation to selected year; hand calc may be at older basis

In short: **Aspen usually reports higher costs** than hand *purchased-only* numbers because it (a) adds design realism (materials, pressure, internals) and (b) often reports **installed** or **bare-module** totals unless you explicitly change the scope.

Example Reconciliations (Show How the Gap Arises)

The following are **illustrative** using your provided sizes. They demonstrate **mechanisms** that make Aspen's values diverge; replace the multipliers with Aspen's actual reports when you run the models.

1) E-301 Methanol Condenser (Area 3,300 m^2)

Hand (purchased only):

$$
C_\text{hand} = 50{,}000\left(\frac{3300}{50}\right)^{0.60} \approx \$0.618\ \text{MM}
$$

Likely Aspen adjustments:

- Material: 304/316SS for methanol/water → **×1.6**
- Pressure rating (condensing side, column pressure + margins) → **×1.15**
- TEMA type upgrade (floating head) + channels → **×1.10**
- Bare-module/installed scope (field labor, piping tie-ins) → **×2.8** (if "Installed")

Illustrative outcome:

$$
C_\text{Aspen,installed}\approx 0.618 \times 1.6 \times 1.15 \times 1.10 \times 2.8
\approx \$3.5\ \text{MM}
$$

Interpretation: Aspen can be **~5–6×** the hand *purchased* number because it's reporting an **installed** stainless floating-head exchanger under pressure.

---

2) T-300 Distillation Column (D = 7.9 m, H = 12.7 m, 23 stages)

Hand (purchased shell + internals, simple 0.6 rule):

$$
C_\text{hand} \approx \$1.053\ \text{MM}
$$

Likely Aspen adjustments:

- Shell thickness at 50 bar service → **×1.3**
- 23 trays/packing + support, downcomers, manways → **×1.4**
- Alloy selection or corrosion allowance (if any) → **×1.2**
- Installed scope (foundations, platforms, ladders, insulation) → **×2.5**

Illustrative outcome:

$$
C_\text{Aspen,installed}\approx 1.053 \times 1.3 \times 1.4 \times 1.2 \times 2.5 \approx \$5.6\ \text{MM}
$$

Interpretation: Trays/platforms/foundations dominate; Aspen's total can substantially exceed the hand shell-only estimate.

---

3) K-300 Recycle Compressor (flow 16,230 L/min)

Hand (flow-based exponent 0.8):
$C_\text{hand} \approx \$0.77\ \text{MM}$

Aspen approach (by power):

1. Compute power from actual **recycle conditions** (polytropic/isentropic with compressibility).
2. Cost the **compressor case** + **driver** (motor/gearbox), baseplate, cooler, controls; apply **installed** factor.

Illustrative: If Aspen calculates 1.2 MW shaft power, a typical installed cost is **$1.5–2.5 MM**, often **2–3×** a simple purchased casing estimate.

Side-by-Side Summary (Using Hand Results from Part a)

Tag	Hand Result (Purchased-Only)*	Aspen Scope Likely	Expected Difference	Primary Causes
E-300	$0.060 MM	Bare-module or Installed	↑	Materials, installation
E-300*	$0.586 MM	Installed	↑↑	Large area, pressure rating, field labor
E-301	$0.618 MM	Installed	↑↑↑	SS, floating head, installation
E-302	$0.896 MM	Installed	↑↑↑	High duty → large area; metallurgy
T-300	$1.053 MM	Installed	↑↑↑	Trays, platforms,

Tag	Hand Result (Purchased-Only)*	Aspen Scope Likely	Expected Difference	Primary Causes
				foundations
D-300/301	$0.132 MM	Installed	↑↑	Thickness, nozzles, internals
K-300	$0.770 MM	Installed	↑↑	Power-based costing, driver & auxiliaries
R-300	$0.600 MM	Installed	↑	HP code vessel, internals, insulation

*Hand results are from Part a (purchased equipment). Aspen entries will be larger if you select **Installed** scope or add design/pressure/material corrections.

Accuracy Classes & What to Report

Estimate Type	Method	AACE Class	Typical Accuracy
Order-of-Magnitude	Hand (capacity rule, Lang factor)	Class 5	**±30–50%**
Study / Budget	Aspen with preliminary design, default factors	Class 4–3	**±15–30%** (if scope consistent)

If the **scope** differs (purchased vs. installed), two accurate numbers can still disagree. Always **align scope and cost year** before comparison.

How to Reconcile in Your Report (Actionable Checklist)

1. **Match Scope:** In Aspen, set reporting to **Purchased** (or convert hand costs to **Installed** using bare-module factors) so both sides reflect the *same scope*.
2. **Align Cost Year:** Use one **cost index** (e.g., CEPCI) for all numbers; escalate hand costs to Aspen's year or vice-versa.
3. **Apply Known Corrections to Hand Costs (if staying purchased-only):**
   - Material factor (f_M), pressure factor (f_P), design/type factor (f_T):
     [

$$C_\text{adj} = C_\text{hand}, f_M, f_P, f_T$$

4. **Document Assumptions:** Metallurgy, pressure, TEMA type, tray count, packing type, driver size.
5. **State Estimate Class & Expected Accuracy** so decision-makers read differences correctly.

Conclusion

- **Hand calculations** provide fast **purchased-equipment** screening values using minimal geometry; they **understate** totals when compared to Aspen **installed** or **bare-module** outputs because they omit metallurgy, pressure-thickness, internals, drivers, and installation.
- **Aspen** typically yields **higher, more realistic** costs once scope is set to installed and detailed design choices are made.
- When **scope and cost year are aligned**, remaining deltas usually arise from **materials, pressure, and internals**—factors you can approximate with simple multipliers in hand calcs to bring both methods within **±15–30%** for a consistent basis.

c) ISBL, OSBL, contingency, engineering costs.

c) ISBL, OSBL, Contingency, and Engineering Costs — Order-of-Magnitude Breakdown

Purpose

Allocate the **Total Installed Capital** for the methanol plant into **Inside Battery Limits (ISBL)**, **Outside Battery Limits (OSBL)**, **Engineering & Home Office (E&H)**, and **Contingency** using industry-standard percentage factors appropriate for a screening (order-of-magnitude) estimate.

Basis & Costing Rules

- **Purchased Equipment Cost (PEC)** from Part (a): **$4.712 MM** (sum of major items R-300, D-300/301, T-300, E-300/301/302, K-300).
- **Estimate class:** AACE Class 5 (±30–50%).
- **Assumed factors (typical for reaction + distillation fluids plants):**
    1. **ISBL installed factor** on PEC: ($F_{\text{ISBL}} = 3.0$)

- Covers setting in place, foundations, piping, electricals, insulation, paint, small instruments, in-unit buildings, and construction labor **within the unit limits**.

2. **OSBL** (utilities, offsites, storage, flare, wastewater tie-ins, site works): **35% of ISBL**.
3. **Engineering & Home Office (E&H)** (process/design, drafting, procurement, construction management): **10% of (ISBL + OSBL)**.
4. **Contingency** (scope growth & minor design development): **8% of (ISBL + OSBL)**.

These selections yield a transparent split that is consistent with typical Lang-factor totals for chemical plants and match the technology mix (reactor + column + recycle + utilities).

Step-by-Step Calculations (with Formulas)

1. **ISBL (Installed in-unit cost)**
   [
   \text{ISBL} = F_{\text{ISBL}}\times \text{PEC}
   = 3.0 \times 4.712 = \boxed{14.136\ \text{MM$}}
   ]
2. **OSBL (Offsites & Utilities)**
   [
   \text{OSBL} = 0.35 \times \text{ISBL}
   = 0.35 \times 14.136 = \boxed{4.9476\ \text{MM$}}
   ]
3. **Direct Field Cost Subtotal**
   [
   D = \text{ISBL} + \text{OSBL}
   = 14.136 + 4.9476 = \boxed{19.0836\ \text{MM$}}
   ]
4. **Engineering & Home Office (E&H)**
   [
   \text{E&H} = 0.10 \times D
   = 0.10 \times 19.0836 = \boxed{1.9084\ \text{MM$}}
   ]

5. **Contingency**

$$
\text{Contingency} = 0.08 \times D
= 0.08 \times 19.0836 = \boxed{1.5267\ \text{MM\$}}
$$

6. **Total Installed Capital (TIC) / Fixed Capital Investment (FCI)**

$$
\text{TIC} = D + \text{E\&H} + \text{Contingency}
= 19.0836 + 1.9084 + 1.5267
= \boxed{22.52\ \text{MM\$}}
$$

This **$22.52 MM** total aligns with an overall Lang factor of ($\approx 22.52/4.712 = 4.78$), which is within the common **4.7–4.8** range for fluids/chemicals plants at this definition level.

Results — Cost Breakdown Table

Category	Basis / Formula	Amount (MM$)	Share of Total
ISBL (installed)	($3.0 \times \text{PEC}$)	**14.136**	**62.8%**
OSBL	($0.35 \times \text{ISBL}$)	**4.948**	**22.0%**
Engineering & Home Office	($0.10 \times (\text{ISBL}+\text{OSBL})$)	**1.908**	**8.5%**
Contingency	($0.08 \times (\text{ISBL}+\text{OSBL})$)	**1.527**	**6.8%**
Total Installed Capital (FCI)	Sum of above	**22.52**	**100%**

Notes on Scope

- **ISBL** includes: reactor R-300, column T-300 (with 23 stages), separators D-300/D-301, exchangers E-300/301/302, recycle compressor K-300, in-unit piping/instruments, in-unit foundations/platforms/insulation.

- **OSBL** covers: cooling-water system capacity, LP steam supply tie-in, electrical substation/MCCs, instrument air, methanol storage/loading, wastewater pre-treatment tie-ins, flare/vent handling, buildings and site work outside unit limits.
- **E&H**: process & mechanical design, procurement, project management, construction supervision, QA/QC, startup assistance (home-office plus field engineering).
- **Contingency** is **not** a reserve for mistakes; it accounts for normal scope growth and minor design development typical of early-stage estimates.

Sensitivity (for transparency)

If you adopt different typical factors, the totals move as follows (PEC fixed at $4.712 MM):

Case	(F_{\text{ISBL}})	OSBL (% of ISBL)	E&H (% of D)	Cont (% of D)	TIC (MM$)
Conservative utilities	3.0	25%	10%	10%	21.43
Base case (reported)	**3.0**	**35%**	**10%**	**8%**	**22.52**
Utility-heavy site	3.0	45%	12%	10%	24.31

This shows the estimate range expected for an order-of-magnitude study.

Conclusion

Using the purchased-equipment total and standard early-phase factors tailored to a reactor–distillation methanol unit, the **capital split** is:

- **ISBL:** $14.14 MM
- **OSBL:** $4.95 MM
- **Engineering & Home Office:** $1.91 MM
- **Contingency:** $1.53 MM
- **Total Installed Capital: $22.52 MM**

This structured breakdown is appropriate for feasibility screening and is consistent with a Lang-factor of ~4.8 for a fluids-processing chemical plant.

d) Total capital cost.

(d) Total Capital Cost — Summary, Method, and Calculation

1. Objective

Report the **Total Capital Cost** (also called **Total Installed Cost / Fixed Capital Investment**) for the methanol plant using an order-of-magnitude approach consistent with standard practice for a fluids chemical unit (reaction + distillation + recycle).

---

2. Costing Basis & Factors

- **Purchased Equipment Cost (PEC)** from the provided equipment list (reactor, column, separators, exchangers, compressor) estimated by capacity-factored hand calculations:
$$\boxed{\text{PEC} = \$4.712\ \text{MM}}$$
- **Inside Battery Limits (ISBL) installed factor on PEC**: $F_{\text{ISBL}} = 3.0$
(covers installation, piping, instruments, foundations, platforms, insulation within unit limits)
- **Outside Battery Limits (OSBL)**: 35% of ISBL
(utilities, storage, flare/vent, wastewater tie-ins, site works, buildings outside unit)
- **Engineering & Home Office (E&H)**: 10% of $(\text{ISBL}+\text{OSBL})$
- **Contingency**: 8% of $(\text{ISBL}+\text{OSBL})$

These factors are typical for AACE Class-5 screening estimates of reaction/separation chemical plants and yield an overall Lang factor ≈ 4.7–4.8.

---

3. Step-by-Step Calculation

3.1 ISBL (Installed in-unit)

$$\text{ISBL} = F_{\text{ISBL}}\times \text{PEC} = 3.0 \times 4.712 = \boxed{14.136\ \text{MM\$}}$$

3.2 OSBL (Offsites & Utilities)

[
\text{OSBL} = 0.35 \times \text{ISBL} = 0.35 \times 14.136 = \boxed{4.9476\ \text{MM$}}
]

3.3 Direct Field Cost Subtotal

[
D = \text{ISBL}+\text{OSBL} = 14.136 + 4.9476 = \boxed{19.0836\ \text{MM$}}
]

3.4 Engineering & Home Office

[
\text{E&H} = 0.10 \times D = 0.10 \times 19.0836 = \boxed{1.9084\ \text{MM$}}
]

3.5 Contingency

[
\text{Contingency} = 0.08 \times D = 0.08 \times 19.0836 = \boxed{1.5267\ \text{MM$}}
]

3.6 Total Capital Cost (TCC / TIC / FCI)

[
\boxed{\text{Total Capital Cost} = D + \text{E&H} + \text{Contingency}
= 19.0836 + 1.9084 + 1.5267
= \mathbf{22.52\ \text{MM$}}}
]

---

4. Capital Breakdown Table

Category	Formula	Amount (MM$)	% of Total
ISBL (installed)	(3.0 \times \text{PEC})	**14.136**	62.8%
OSBL	(0.35 \times \text{ISBL})	**4.948**	22.0%
Engineering & Home Office	(0.10 \times (\text{ISBL}+ \text{OSBL}))	**1.908**	8.5%
Contingency	(0.08 \times (\text{ISBL}+ \text{OSBL}))	**1.527**	6.8%

Category	Formula	Amount (MM$)	% of Total
Total Capital Cost	Sum	**22.52**	**100%**

5. Notes on Accuracy & Sensitivity

- **Estimate Class:** AACE Class-5 (order-of-magnitude), expected accuracy **±30–50%**.
- If your site is **utility-heavy** (e.g., significant new CW/steam generation, storage), raising OSBL to 45% and E&HO to 12% would move TCC toward **$24–25 MM**. Conversely, a brownfield site with existing utilities (OSBL ~25%) would reduce TCC toward **$21–22 MM**.

Reported Total Capital Cost (base case): ($\mathbf{\$22.5\ million}$).

Variable cost of production (raw materials and utilities) forecasted prices for raw materials and products.

(e) Variable Cost of Production — Raw Materials & Utilities

(with current market price references and worked examples)

1) Objective

Estimate the **variable cost of production (VCOP)** for methanol based on **raw materials** (H_2, CO_2) and **utilities** (electricity, steam, cooling water), and provide **current market price benchmarks/forecasts** for these inputs and for the methanol product.

---

2) Chemical Basis (per metric ton of methanol)

Reaction: $CO_2 + 3\ H_2 \rightarrow CH_3OH + H_2O$

From stoichiometry (per **1.000 t MeOH**, assuming net consumption equals stoichiometric requirements; recycle/inefficiencies can be added later as factors):

- **Hydrogen**: ($n_{\mathrm{H_2}} = 3\ \text{mol} \Rightarrow m_{\mathrm{H_2}}/m_{\mathrm{MeOH}} = 6/32 = 0.1875$)
  → **0.1875 t H_2 per t MeOH** (i.e., **187.5 kg H_2/t MeOH**)
- **CO_2**: ($44/32 = 1.375$)
  → **1.375 t CO_2 per t MeOH**

- **Water byproduct**: (18/32 = 0.5625) (no cost credit assumed)

If your plant achieves <100% single-pass conversion with recycle, the **net** consumption remains stoichiometric; only **utility** usage changes (more compression/heating/cooling).

---

3) Market Price Benchmarks (Nov 2025)

Item	Benchmark (unit)	Notes & Sources
Methanol (product)	**$802/MT** (U.S. Gulf Coast, Nov 2025 posted reference)	Methanex North America Non-Discounted Reference, posted Oct 28, 2025. (Methanex)
Hydrogen, gray (SMR, gate price)	**$1–$3/kg** (use **$1.6/kg** base; scenario range shown)	Range compiled from industry summaries and reviews in 2025. (stargatehydrogen.com)
Hydrogen, green (electrolysis)	**$4–$12/kg** (use **$5/kg** scenario)	Cost range in 2025 and DOE cost-reduction targets context. (Energies Media)
CO_2, bulk liquid (industrial/food grade)	**$267–$590/MT** (regional; use **$600/MT** conservative)	Recent U.S. reports for Q2–Q3 2025 show ~$590/MT DEL IL; price varies widely by region/purity. (ChemAnalyst)
CO_2 policy credit (45Q, utilization/sequestration)	**Policy-dependent; up to $85/MT** (utilization/sequestration)	Credit value depends on project vintage/qualifications; recent guidance/bills discuss $60–$85+/t ranges. Use **$60–$85/t** for utilization/sequestration *if eligible*. (Reuters)
Electricity (industrial retail)	**~8–10¢/kWh** (U.S. industrial avg, Aug 2025)	EIA Electric Power Monthly (Table 5.3, 5.6A); industrial prices vary by state. (U.S. Energy Information Administration)
Natural gas (Henry Hub)	**$3.0–$3.5/MMBtu** (Fall 2025 spot; 12-mo strip ~$4.05)	Use for steam generation costing; adjust for boiler efficiency. (U.S. Energy Information Administration)

Use your site's actual contract prices if available; the table above provides **defensible public benchmarks** for a feasibility-level estimate.

---

4) Utilities Basis (set from your Utilities Table)

Please plug in the specific duties/flows from your provided **Utilities Table**. If not yet finalized, the following **template** (and one **illustrative** set) demonstrates the method:

Utility	Symbol	Unit Intensity (per t MeOH)	Unit Price	Formula	Cost (USD/t MeOH)
Electricity	(E)	$(E_{\text{kWh/t}})$ kWh/t	(P_e) \$/kWh	$(E_{\text{kWh/t}} \times P_e)$	—
LP Steam (from NG)	(S)	$(S_{\text{GJ/t}})$ GJ/t	(P_s) \$/GJ	$(S_{\text{GJ/t}} \times P_s)$	—
Cooling Water	(CW)	$(V_{\text{m}^3/\text{t}})$ m³/t	(P_{cw}) \$/m³	$(V_{\text{m}^3/\text{t}} \times P_{cw})$	—

Typical costing linkages:

- $P_s \approx \dfrac{P_{\text{NG}}}{\eta_{\text{boiler}}\times 1.055}$ \$/GJ (with P_{NG} in \$/MMBtu; η ≈ 0.85–0.90).
- Cooling water unit prices are site-specific (often \$0.03–\$0.10/m³ all-in).

Illustrative numbers (for a complete worked example below; replace with your table values):
$E_{\text{kWh/t}}=60$, $P_e=\$0.09/\text{kWh}$; $S_{\text{GJ/t}}=2.0$, $P_{\text{NG}}=\$3.20/\text{MMBtu}$, $\eta=0.85 \Rightarrow P_s\approx \$3.56/\text{GJ}$; $V_{m^3/t}=20$, $P_{cw}=\$0.06/\text{m}^3$. ([U.S. Energy Information Administration](#))

---

5) Worked Examples — Variable Cost per t MeOH

5.1 Raw-Material Consumption (fixed by stoichiometry)

- **H_2: 187.5 kg/t MeOH**
- **CO_2: 1.375 t/t MeOH**

5.2 Utility Cost (using illustrative intensities above)

- Electricity: (60\ \text{kWh/t} \times \$0.09/\text{kWh} = \$5.40)
- LP Steam: (2.0\ \text{GJ/t} \times \$3.56/\text{GJ} = \$7.12)
- Cooling water: (20\ \text{m}^3/\text{t} \times \$0.06/\text{m}^3 = \$1.20)
  Utilities subtotal (illustrative): \$13.72/t MeOH

5.3 Scenario Table (all values USD per metric ton methanol)

Scenario	H_2 Price	CO_2 Price/Credit	Raw-Mat Cost	Utilities*	VCOP
A. Gray H_2 + Purchased CO_2	\$1.60/kg	**\$600/t (cost)**	H_2: (187.5\times1.60 = \$300) + CO_2: (1.375\times600 = \$825) → **\$1,125**	\$13.72	**\$1,138.72**
B. Gray H_2 + CO_2 free	\$1.60/kg	**\$0/t**	**\$300**	\$13.72	**\$313.72**
C. Green H_2 + 45Q Utilization Credit	\$5.00/kg	**(-\$60/t) credit**	H_2: (187.5\times5.00 = \$937.5) + CO_2 credit: (1.375\times60 = -\$82.5) → **\$855.0**	\$13.72	**\$868.72**
D. Blue H_2 + CO_2 free	\$2.50/kg	**\$0/t**	H_2: (187.5\times2.50 = \$468.75) → **\$468.75**	\$13.72	**\$482.47**

* Utilities shown with the illustrative intensities/prices above; replace with your **Utilities Table** values to finalize VCOP.
CO_2 policy credits depend on project qualification/eligibility; use \$60–\$85/t if the plant is eligible under current law/bills. (Reuters)

---

6) Product Price Context (Nov 2025)

- **Methanol reference price** (North America, posted): **\$802/MT**. (Methanex)

Illustration (gross margin, not including fixed OPEX or depreciation):

Scenario	VCOP (from §5.3)	Reference Methanol Price	Indicative Gross Margin
A. Gray + Purchased CO_2	\$1,139/t	\$802/t	**–\$337/t**
B. Gray + CO_2 free	\$314/t	\$802/t	**+\$488/t**
C. Green + 45Q credit	\$869/t	\$802/t	**–\$67/t**

Scenario	VCOP (from §5.3)	Reference Methanol Price	Indicative Gross Margin
D. Blue + CO_2 free	\$482/t	\$802/t	**+\$320/t**

The **CO_2 cost/credit** and **H_2 pathway** dominate VCOP. **Utilities** are comparatively small at this scale unless compression/recycle duties are very large.

---

7) How to Finalize with Your Data

1. **Insert your Utilities Table numbers** (E_{\text{kWh/t}}, S_{\text{GJ/t}}, V_{\text{m}^3/\text{t}}) and site prices.
2. **Replace market benchmarks** with your **actual contracts** if available.
3. If you have **single-pass conversion, purge, or higher recycle compression**, adjust **utility intensities** accordingly.
4. If pursuing **45Q**, confirm eligibility and applicable credit value with current tax counsel/regulatory guidance before treating credits as variable offsets. (Legal Information Institute)

---

8) Summary

- **Stoichiometric consumption per t MeOH: 0.1875 t H_2** and **1.375 t CO_2**.
- **Market benchmarks (Nov 2025):** Methanol **\$802/t**; gray H_2 **\$1–3/kg** (use **\$1.6/kg**), green H_2 **\$4–12/kg** (scenario **\$5/kg**); CO_2 bulk **\$267–\$590/t** (scenario **\$600/t**); industrial power ~**8–10¢/kWh**; Henry Hub ~**\$3–3.5/MMBtu**. (Methanex)
- **VCOP is highly sensitive to H_2 & CO_2 terms**. With **purchased CO_2** and **green H_2**, VCOP can exceed product price; with **gray/blue H_2** and **free (or credited) CO_2**, VCOP can be substantially below product price.

Replace the illustrative utility intensities with your **provided utilities table** to produce your final VCOP and proceed to the cash-costs and profitability sections.

Fixed cost of production (assume that a year equals 8000 hours. This is about 330 days, which allows for periodic shut-down and maintenance)

(f) Fixed Cost of Production (FCOP)

Basis: 8,000 operating hours per year (≈330 days). Results shown **per year** and **per operating hour**. Use the final line to compute **$/t** once your annual production rate is known.

---

1) Costing Basis & Definitions

Fixed costs do not scale with short-term throughput. Included here:

- Operating labor (shift operators)
- Supervision
- Maintenance (labor + materials)
- Operating supplies (consumables not tied to tonnage)
- QA/QC & laboratory
- Plant overhead (shop services, utilities for buildings, maintenance support, etc.)
- Taxes & insurance (on fixed assets)
- Administration (G&A)

Given/derived constants for this estimate

- Operating hours per year: **8,000 h/y**
- Fixed Capital Investment (from earlier section): **FCI = $22.52 MM**
- Selected industry-typical factors (screening estimate):
    - Supervision = **15% of operating labor**
    - Maintenance (total) = **3% of FCI**
        - Split: **30% labor / 70% materials**
    - Operating supplies = **10% of maintenance**
    - QA/QC & lab = **20% of operating labor**
    - Plant overhead = **60% of (operating labor + maintenance labor)**
    - Taxes & insurance = **1.5% of FCI**
    - Administration (G&A) = **25% of (operating labor + maintenance labor + supervision + QA/QC)**

Staffing & wage (illustrative, adjust to your site):

- **Operators per shift position (OPS) = 5**
- **Loaded hourly wage** (wages + benefits) = **$43 / h**

You can re-run the same equations with your own OPS and loaded wage if different.

---

2) Formulas

Let:

- (N_\text{ops}) = operators per shift position (here 5)
- (w_L) = loaded wage ($/h) (here 43)
- (H) = hours/year (here 8,000)
- (\text{FCI}) = $22.52 MM

Operating labor
[
C_{\text{OL}} = N_\text{ops}, w_L, H
]

Supervision
[
C_{\text{SUP}} = 0.15, C_{\text{OL}}
]

Maintenance (total)
[
C_{\text{MAIN}} = 0.03, \text{FCI},\quad
C_{\text{MAIN,L}} = 0.30, C_{\text{MAIN}},\quad
C_{\text{MAIN,M}} = 0.70, C_{\text{MAIN}}
]

Operating supplies
[
C_{\text{SUPPL}} = 0.10, C_{\text{MAIN}}
]

QA/QC & Lab
[

C_{\text{QA}} = 0.20, C_{\text{OL}}
]

Plant overhead
[
C_{\text{POH}} = 0.60,(C_{\text{OL}} + C_{\text{MAIN,L}})
]

Taxes & insurance
[
C_{\text{T&I}} = 0.015, \text{FCI}
]

Administration (G&A)
[
C_{\text{G&A}} = 0.25, (C_{\text{OL}} + C_{\text{MAIN,L}} + C_{\text{SUP}} + C_{\text{QA}})
]

Total Fixed Cost of Production (annual)
[
\boxed{ \text{FCOP}\text{y} = C{\text{OL}} + C_{\text{SUP}} + C_{\text{MAIN,L}} + C_{\text{MAIN,M}} + C_{\text{SUPPL}} + C_{\text{QA}} + C_{\text{POH}} + C_{\text{T&I}} + C_{\text{G&A}} }
]

Per operating hour
[
\boxed{ \text{FCOP}\text{h} = \dfrac{\text{FCOP}\text{y}}{H} }
]

Per metric ton of product (once annual production (P_\text{y}) t/y is known)
[
\boxed{ \text{FCOP}{$/t} = \dfrac{\text{FCOP}\text{y}}{P_\text{y}} }
]

---

3) Worked Example (with the values above)

Component	**Formula (from §2)**	**Annual Cost (USD)**
Operating labor	(5 \times 43 \times 8000)	**1,720,000**

Component	Formula (from §2)	Annual Cost (USD)
Supervision (15% OL)	(0.15 \times 1{,}720{,}000)	**258,000**
Maintenance — total (3% FCI)	(0.03 \times 22.52\text{MM})	**675,600**
└─ Labor (30%)	(0.30 \times 675{,}600)	**202,680**
└─ Materials (70%)	(0.70 \times 675{,}600)	**472,920**
Operating supplies (10% maint.)	(0.10 \times 675{,}600)	**67,560**
QA/QC & lab (20% OL)	(0.20 \times 1{,}720{,}000)	**344,000**
Plant overhead (60% of OL + maint. labor)	(0.60 \times (1{,}720{,}000 + 202{,}680))	**1,153,608**
Taxes & insurance (1.5% FCI)	(0.015 \times 22.52\text{MM})	**337,800**
Administration (25% of OL + maint. labor + supervision + QA)	(0.25 \times (1{,}720{,}000 + 202{,}680 + 258{,}000 + 344{,}000))	**631,170**
Total Fixed Cost (FCOP_y)	Sum of above	**$5,187,738**
Per operating hour (FCOP_h)	(5{,}187{,}738 / 8{,}000)	**$648.47 / h**

To convert to **$/t**, divide $5,187,738 by your plant's annual production (P_\text{y}) (t/y).
Example: If (P_\text{y}=100{,}000\ \text{t/y}) → **$51.88/t** fixed cost.

---

4) How to Tailor This to Your Plant

1. Replace **OPS** and **loaded wage** with your staffing plan and HR rates.
2. If your corporate standards use different **factors** (e.g., maintenance 2–5% of FCI, T&I 1–2% of FCI, plant overhead 50–80% of direct labor), re-run the same formulas.
3. Once you provide **annual nameplate production**, compute **FCOP $/t** directly using the last equation.

Reported results (with the example assumptions):

- **Annual FCOP: $5.19 MM/y**
- **FCOP per hour: $648.47/h**
- **FCOP per ton: $5.19 MM / (P_\text{y})** (enter your (P_\text{y})).

Revenues and gross profit.

(g) Revenues and Gross Profit

1) Definitions & What You Will Report

- **Annual Revenue, (R_y)**: money earned from selling methanol.
  [
  R_y = P_{\text{MeOH}};[$/t]\times Q_y;[t/y]
  ]
- **Gross Profit (process-econ convention for this course)**: **Revenue – Variable Cost of Production (VCOP)**. Fixed costs are handled in a later "operating profit" step.
  [
  \text{Gross Profit}*y = (P*{\text{MeOH}} - \text{VCOP}*{$/t}) \times Q_y*
  ]
  [
  \text{Gross Margin per ton} = P{\text{MeOH}} - \text{VCOP}_{$/t}
  ]

If your instructor defines *gross profit* as Revenue – COGS including fixed factory charges, replace (\text{VCOP}*{$/t}) with (\text{COGS}*{$/t}) in the same equations.

---

2) Inputs (to be filled from your case)

Symbol	Description	Units	Value (enter yours)
(Q_y)	Annual methanol production	t/y	☐
(P_{\text{MeOH}})	Selling price of methanol	$/t	☐
(\text{VCOP}_{$/t})	Variable cost per ton (raw materials + utilities)	$/t	☐

3) Calculation Formulas (ready to paste)

Annual Revenue
[
R_y = P_{\text{MeOH}} \times Q_y
]

Gross Margin per ton
[

\text{GM}{$/t} = P{\text{MeOH}} - \text{VCOP}_{$/t}
]

Annual Gross Profit
[
\text{GP}y = \text{GM}{$/t} \times Q_y
]

---

4) Worked Example (illustrative numbers)

Replace the three gray cells below with your own values; the math flows automatically.

Assumptions (illustrative):
(Q_y = 100{,}000\ \text{t/y}), (P_{\text{MeOH}} = $743/t), (\text{VCOP}_{$/t} = $500/t).

4.1 Revenue

[
R_y = 743 \times 100{,}000 = \boxed{$74{,}300{,}000/y}
]

4.2 Gross Margin per ton

[
\text{GM}_{$/t} = 743 - 500 = \boxed{$243/t}
]

4.3 Annual Gross Profit

[
\text{GP}_y = 243 \times 100{,}000 = \boxed{$24{,}300{,}000/y}
]

---

5) Sensitivity Table (useful for your report)

Hold (Q_y) fixed and vary either price or VCOP to see gross profit swing.

Case: (Q_y = 100{,}000\ \text{t/y})

Methanol Price ($/t)	VCOP ($/t)	Gross Margin ($/t)	Annual Gross Profit ($/y)
700	500	200	20,000,000
743	500	243	24,300,000

Methanol Price ($/t)	VCOP ($/t)	Gross Margin ($/t)	Annual Gross Profit ($/y)
800	500	300	30,000,000
743	550	193	19,300,000
743	600	143	14,300,000

To update, plug in **your** (Q_y), (P_{MeOH}), and ($\text{VCOP}_{\$/t}$). The same equations apply if you later include by-product credits—subtract those from VCOP first, then compute margin.

---

6) What to Include in Your Final Report Section

- A one-line statement of **annual revenue** at your chosen selling price and capacity.
- A table showing **per-ton gross margin** and **annual gross profit** at your **base case** and at ±10% price and ±10% VCOP for robustness.
- A note clarifying the definition you used for **gross profit** (process-econ convention: excludes fixed costs).

Deliverables (this answer): formulas, example calculations, and a sensitivity table formatted for direct inclusion in your report.

h) Simple pay-back period and ROI.

(h) Simple Pay-Back Period and ROI

1) Objective

Compute the **simple pay-back period (PBP)** and **return on investment (ROI)** for the methanol plant using a transparent, screening-level convention.

---

2) Definitions & Conventions (screening level)

- **Annual Revenue**: ($R_y = P_{\text{MeOH}}[\$/t]\times Q_y[\text{t/y}]$)
- **Gross Profit (GP(_y))**: ($(P_{\text{MeOH}}-\text{VCOP}_{\$/t}) \times Q_y$)
- **Operating Profit before depreciation, interest, tax** (a proxy for simple *annual cash flow* at this level):
$$\boxed{\text{OP}_y = \text{GP}_y - \text{FCOP}_y}$$

- **Simple Pay-Back Period** (years):
$$
\boxed{\text{PBP} = \dfrac{\text{FCI}}{\text{OP}_y}}
$$
- **Simple ROI** (percent, on fixed capital):
$$
\boxed{\text{ROI} = \dfrac{\text{OP}_y}{\text{FCI}}\times 100\% }
$$

Notes:

1. “Simple” PBP/ROI ignore financing, taxes, and depreciation.
2. If your course defines ROI differently (e.g., using net profit after fixed costs and depreciation), swap in that annual figure in the same formulas.

---

3) Inputs (from prior parts of this assignment)

Symbol	Description	Value
Q_y	Annual methanol production	**100,000 t/y** *(illustrative for the worked example—replace with your value if different)*
P_{MeOH}	Selling price	**\$743 / t**
$\text{VCOP}_{\$/t}$	Variable cost per ton	**\$500 / t**
FCOP_y	Fixed cost of production (annual)	**\$5,187,738 / y**
FCI	Fixed Capital Investment (Total capital cost)	**\$22.52 MM**

4) Worked Example (base case)

4.1 Annual Revenue

$$
R_y = 743 \times 100{,}000 = \mathbf{\$74{,}300{,}000 / y}
$$

4.2 Gross Profit

[
\text{GP}_y = (743-500)\times 100{,}000 = 243 \times 100{,}000
= \mathbf{$24{,}300{,}000 / y}
]

4.3 Operating Profit (proxy cash flow)

[
\text{OP}_y = \text{GP}_y - \text{FCOP}_y
= 24{,}300{,}000 - 5{,}187{,}738
= \mathbf{$19{,}112{,}262 / y}
]

4.4 Simple Pay-Back Period

[
\text{PBP} = \dfrac{22.52\ \text{MM}}{19.112262\ \text{MM/y}}
= \mathbf{1.178\ \text{y}} ;;(\approx \mathbf{14.14\ \text{months}})
]

4.5 Simple ROI

[
\text{ROI} = \dfrac{19.112262}{22.52}\times 100%
= \mathbf{84.87%}
]

---

5) Sensitivity (price & VCOP swings; (Q_y=100{,}000\ \text{t/y}), (\text{FCOP}_y=$5{,}187{,}738), FCI=$22.52MM)

Case	Methanol Price ($/t)	VCOP ($/t)	Gross Margin ($/t)	OP(_y) ($/y)	PBP (y)	ROI (%)
Base	743	500	243	**19,112,262**	**1.178**	**84.87**
Low price	700	500	200	14,812,262	1.520	65.77
High price	800	500	300	24,812,262	0.908	110.18
Higher VCOP	743	550	193	14,112,262	1.596	62.67
Much higher VCOP	743	600	143	9,112,262	2.471	40.46

6) How to finalize for your report

1. Replace the **three commercial inputs** (Q_y), (P_{MeOH}), and ($\text{VCOP}_{\$/t}$) with your case values.
2. Keep (FCOP_y) and **FCI** from your earlier sections (or update if you refined them).
3. Recompute §4.3–§4.5 with the same equations.
4. Include this sensitivity table (± price, ± VCOP) to show robustness.

Reported (base-case) results:
Simple PBP = 1.18 years (≈14.1 months); **Simple ROI = 84.9%**.

NPV value calculated at a discount rate of 14%, using MACRS depreciation with a 7- year recovery term, and 5-year recovery term. Compare and discuss the difference.

(i) NPV at 14% with MACRS 7-yr vs 5-yr — Calculation & Comparison

1) Objective

Compute **project NPV at a 14% discount rate** twice—first with **MACRS 7-year** and then with **MACRS 5-year** depreciation—then compare the results and explain the difference.

---

2) Basis & Assumptions (consistent with earlier parts of this assignment)

- **Discount rate:** ($i = 14\%$)
- **Project life: 10 operating years** (Years 1–10), no salvage value, no working capital modeled.
- **Tax rate: 35%** (course convention).
- **Capex at ($t=0$): FCI = $22.52 MM**.
- **Annual operations (Years 1–10):**
  - Annual production ($Q_y = 100{,}000\ \text{t/y}$)
  - Methanol price ($P_{\text{MeOH}} = \$743/\text{t}$)
  - Variable cost ($\text{VCOP}_{\$/t} = \$500/\text{t}$)
  - Fixed cost (annual) ($\text{FCOP}_y = \$5{,}187{,}738$)
- These give **gross profit** and **EBIT before depreciation**:
  [

\text{GP}y=(743-500)\times 100{,}000=$24{,}300{,}000/y\quad\Rightarrow\quad
\text{EBIT}{\text{pre-dep}}=24{,}300{,}000-5{,}187{,}738=\mathbf{$19{,}112{,}
262/y}
]

You can substitute your own (Q_y, P_{\text{MeOH}}, \text{VCOP}_{$/t}, \text{FCOP}_y) and re-run the same steps.

---

3) Depreciation Schedules (MACRS, half-year convention)

5-year MACRS (% of capital per year):
20.00, 32.00, 19.20, 11.52, 11.52, 5.76

7-year MACRS (% of capital per year):
14.29, 24.49, 17.49, 12.49, 8.93, 8.92, 8.93, 4.46

Depreciation in year (t): (\text{Dep}_t = \text{FCI}\times f_t).

---

4) Cash-Flow Method (per year (t=1\ldots 10))

1. **EBIT**: (\text{EBIT}t=\text{EBIT}{\text{pre-dep}}-\text{Dep}_t)
2. **Taxes**: (\text{Tax}_t = 0.35\times \text{EBIT}_t)
3. **Net income**: (\text{NI}_t=\text{EBIT}_t-\text{Tax}_t)
4. **After-tax operating cash flow**:
   [
   \boxed{\text{CF}_t=\text{NI}_t+\text{Dep}t
   = \text{EBIT}{\text{pre-dep}},(1-0.35)+0.35,\text{Dep}_t}
   ]
5. **NPV**: (\displaystyle \text{NPV}=\sum_{t=0}^{10}\frac{\text{CF}_t}{(1+0.14)^t}), with (\text{CF}_0=-\text{FCI}).

---

5) Key Yearly Numbers (selected rows)

5.1 5-Year MACRS (FCI = $22.52 MM)

Year	Depreciation ($)	EBIT ($)	Tax ($)	Net Income ($)	Cash Flow ($)
0	—	—	—	—	**–22,520,000**
1	4,504,000	14,608,262	5,112,892	9,495,370	**13,999,370**
2	7,206,400	11,905,862	4,167,052	7,738,810	**14,945,210**
3	4,323,840	14,788,422	5,175,948	9,612,474	**13,936,314**
4	2,594,304	16,518, - actually 19,112,262-2,594,304=16,518, -	—	—	**13,330,977**
…	…	…	…	…	…
9	0	19,112,262	6,689,292	12,422,970	**12,422,970**
10	0	19,112,262	6,689,292	12,422,970	**12,422,970**

(Full-year calculations use the formulas in §4; minor rounding occurs.)

5.2 7-Year MACRS (FCI = $22.52 MM)

Year	Depreciation ($)	EBIT ($)	Tax ($)	Net Income ($)	Cash Flow ($)
0	—	—	—	—	**–22,520,000**
1	3,218,108	15,894,154	5,562, -	—	**13,549,308**
2	5,515,148	13,597,114	4,758, -	—	**14,353,272**
3	3,938,748	15,173,514	5,310, -	—	**13,801,532**
…	…	…	…	…	…
9	0	19,112,262	6,689,292	12,422,970	**12,422,970**
10	0	19,112,262	6,689,292	12,422,970	**12,422,970**

Interpretation: 5-yr MACRS front-loads more depreciation (larger early tax shields) than 7-yr MACRS, so its early-year cash flows are slightly higher.

---

6) Results — NPV at 14%

Case	NPV @ 14% ($)
MACRS 5-year	**47,840,743**

Case	NPV @ 14% ($)
MACRS 7-year	47,356,740
Difference (5-yr – 7-yr)	+484,003

7) Discussion — Why 5-yr > 7-yr (and by how much)

1. **Timing of tax shields:** With the same total depreciable basis, the **5-year schedule** recognizes a **larger share of depreciation earlier**, creating bigger **early tax shields** ((0.35\times \text{Dep})). Because cash today is worth more than cash later (14% discount), **front-loaded shields raise NPV**.
2. **Magnitude:** In this base case, the NPV uplift is **about $0.48 MM (~1.0%)**—noticeable but not dominant, because operations are very profitable either way.
3. **When the gap grows:** Higher tax rates, higher discount rates, larger FCI, or shorter project lives **increase** the benefit of faster depreciation.
4. **When it shrinks:** If the plant's taxable income is low (or negative) in early years—e.g., high VCOP or lower prices—the immediate shield may be partially unused (NOL carry), reducing the 5-yr advantage.

---

8) How to replicate with your numbers

- Keep §4 formulas, plug in **your** (Q_y), price, VCOP, FCOP, FCI, and pick the MACRS table (5-yr or 7-yr).
- Extend or shorten the project life if specified by your instructor (the method is the same).
- If you include **working capital** or **salvage**, add those cash flows in Year 0 and Year (N), respectively, before discounting.

Bottom line: At 14% discount, **MACRS 5-year** yields a **higher NPV** than **7-year** because it accelerates the tax shield into earlier years.

Discounted cash flow rate of return using MACRS depreciation with a 7-year recovery term, and 5-year recovery term. Compare these

(j) Discounted Cash-Flow Rate of Return (IRR) with MACRS 7-yr vs 5-yr

1) Objective

Compute the **DCF rate of return (IRR)** for the methanol project twice—once using **MACRS 7-year** and once using **MACRS 5-year** depreciation—and compare the results.

---

2) Basis & Inputs (consistent with earlier parts)

- **Fixed Capital Investment (FCI) = $22.52 MM** at (t=0)
- **Project life:** 10 operating years (Years 1–10), no salvage, no working capital modeled
- **Tax rate:** 35%
- **Discount rate (for context/NPV checks):** 14% (not used to compute IRR, but shown for verification)
- **Annual operations (Years 1–10):**
    - EBIT **before depreciation** (from revenue – VCOP – FCOP): (\mathbf{19{,}112{,}262\ $/y})

Cash-flow identity (each operating year (t))
[
\text{CF}*t=\underbrace{(\text{EBIT}*{\text{pre-dep}}-\text{Dep}*t)}*{\text{taxable}},(1-0.35)+\text{Dep}*t*
=\text{EBIT}{\text{pre-dep}},(1-0.35)+0.35,\text{Dep}_t
]
Initial cash flow: (\text{CF}_0=-\text{FCI}).

---

3) Depreciation Schedules (MACRS, half-year convention)

5-year MACRS (% of basis by year 1→6): 20.00, 32.00, 19.20, 11.52, 11.52, 5.76
7-year MACRS (% of basis by year 1→8): 14.29, 24.49, 17.49, 12.49, 8.93, 8.92, 8.93, 4.46

[
\text{Dep}_t = 22.52\text{ MM}\times f_t
]

---

4) Worked Example — Annual Cash Flows

4.1 Example calculation (Year 1, 5-yr MACRS)

[
\begin{aligned}

\text{Dep}_1 &= 0.20\times 22.52 = 4.504\ \text{MM} \
\text{CF}_1 &= 19.112262(1-0.35)+0.35(4.504) \
&= 12.423 + 1.576 \
&= \mathbf{13.999\ \text{MM}}
\end{aligned}
]

4.2 Cash-flow table — 5-year MACRS

Year	Depreciation (MM$)	Cash Flow (MM$)
0	—	**–22.520**
1	4.504	**13.999**
2	7.206	**14.945**
3	4.323	**13.936**
4	2.594	**13.331**
5	2.594	**13.331**
6	1.297	**12.877**
7	0	**12.423**
8	0	**12.423**
9	0	**12.423**
10	0	**12.423**

4.3 Cash-flow table — 7-year MACRS

Year	Depreciation (MM$)	Cash Flow (MM$)
0	—	**–22.520**
1	3.219	**13.549**
2	5.516	**14.353**
3	3.939	**13.802**
4	2.813	**13.407**
5	2.011	**13.127**

Year	Depreciation (MM$)	Cash Flow (MM$)
6	2.009	**13.126**
7	2.011	**13.127**
8	1.004	**12.775**
9	0	**12.423**
10	0	**12.423**

(Minor rounding; totals are consistent with the formulas above.)

---

5) Results — IRR (DCF Rate of Return)

Depreciation Method	IRR (DCF ROR)
MACRS 5-year	**61.66%**
MACRS 7-year	**60.24%**
Difference (5-yr – 7-yr)	**+1.42 percentage points**

Check (optional): using the same cash flows, NPV at 14% equals **$47.84 MM (5-yr)** vs **$47.36 MM (7-yr)**—the faster schedule gives a slightly higher NPV for the same reason it gives a higher IRR (earlier tax shields).

---

6) Discussion — Why 5-yr > 7-yr

1. **Front-loaded tax shield:** 5-year MACRS recognizes more depreciation in early years, increasing **early after-tax cash flows**. Because IRR is sensitive to **timing**, earlier cash boosts IRR.
2. **Magnitude:** With strong operating earnings, the uplift is modest (≈ **+1.4 pp**). The gap widens if: tax rate is higher, FCI is larger relative to operating cash, discount rate is higher, or project life is shorter.
3. **If early profits were thin:** The immediate shield might be partly unusable (NOLs), reducing the benefit of 5-year acceleration.

---

7) Conclusion

On this basis, the project's DCF ROR is **~61.7% with 5-year MACRS** versus **~60.2% with 7-year MACRS**. The **accelerated (5-yr) schedule** raises returns by bringing the tax shield forward, but the effect is incremental because operating cash flows are already strong.

Sensitivity Analysis on capital cost, utilities, raw material cost and product cost. Provide a detailed explanation.

(k) Sensitivity Analysis — Capital Cost, Utilities, Raw Materials, and Product Price

1) Objective

Quantify how **capital cost (FCI)**, **utilities**, **raw materials**, and **product price** affect project economics. We report impacts on:

- **Operating profit** (proxy annual cash flow at this course level)
- **Simple pay-back (PBP)** and **ROI**
- **NPV change** at **14%** (approximate, holding depreciation schedule fixed)

---

2) Base Case (consistent with earlier parts)

Item	**Value**	**Notes**
Annual production, (Q_y)	**100,000 t/y**	
Methanol selling price, (P)	**$743/t**	
VCOP per ton	**$500/t**	split as **Raw = 486.28**, **Utilities = 13.72**
FCOP (annual)	**$5,187,738 /y**	
Fixed Capital Investment, **FCI**	**$22.52 MM**	
Tax rate	**35%**	for NPV deltas
Discount rate for NPV	**14%**	
Project life for NPV	**10 years**	no salvage, no WC

Derived base metrics

[
\begin{aligned}
\text{Gross margin} &= (743-500)=\mathbf{243\ $/t}\
\text{Gross profit} &= 243\times100{,}000=\mathbf{24.30\ MM$/y}\
\text{Operating profit (OP}_y) &= 24.30-5.188=\mathbf{19.112\ MM$/y}\

\text{PBP} &= \dfrac{22.52}{19.112}=\mathbf{1.178\ y}\
\text{ROI} &= \dfrac{19.112}{22.52}=\mathbf{84.9%}
\end{aligned}
]

NPV delta approximation used for sensitivities
A small change in price or costs shifts **EBIT (pre-depr.)** by (\Delta \text{EBIT}). Because depreciation is unchanged, the **after-tax cash-flow change** is:
[
\Delta \text{CF} \approx (1-0.35),\Delta \text{EBIT}=0.65,\Delta\text{EBIT}
]
The present-worth factor at 14% for 10 years is:
[
\text{PWF}*{10,14%}=\frac{1-(1+0.14)^{-10}}{0.14}=\mathbf{5.216}*
]
So:
[
\boxed{\Delta \text{NPV} \approx 0.65 \times \Delta\text{EBIT} \times 5.216 ;=; \mathbf{3.39}\times \Delta\text{EBIT}}
]
with (\Delta\text{EBIT} = Q_y(\Delta P - \Delta \text{VCOP}{$/t})).
For **capital cost**, (\Delta \text{NPV} \approx -\Delta \text{FCI}) (ignoring the small change to tax shields; this is a conservative one-for-one effect).

---

3) Scenario Results (±10% and ±20%)

3.1 Product Price Sensitivity (holds VCOP, FCOP, FCI fixed)

Scenario	OP(_y) (MM$)	PBP (y)	ROI (%)	ΔEBIT (MM$/y)	Approx ΔNPV (MM$)
Base	19.112	1.178	84.9	0.000	0.000
+10% price (→ $817.3/t)	26.542	0.848	117.9	+7.430	**+25.19**
–10% price (→ $668.7/t)	11.682	1.928	51.9	–7.430	**–25.19**
+20% price (→ $891.6/t)	33.972	0.663	150.9	+14.860	**+50.38**

Scenario	OP(_y) (MM$)	PBP (y)	ROI (%)	ΔEBIT (MM$/y)	Approx ΔNPV (MM$)
–20% price (→ $594.4/t)	4.252	5.296	18.9	–14.860	**–50.38**

Example calc (+10% price)
(\Delta P = +74.3\ $/t \Rightarrow \Delta\text{EBIT}=100{,}000\times74.3 = 7.43\ \text{MM$/y}).
(\Delta\text{NPV}\approx 3.39\times 7.43 = \mathbf{25.2\ MM$}).

3.2 Raw-Material Cost Sensitivity (holds price, utilities, FCOP, FCI fixed)

Scenario	OP(_y) (MM$)	PBP (y)	ROI (%)	ΔEBIT (MM$/y)	Approx ΔNPV (MM$)
Base	19.112	1.178	84.9	0.000	0.000
+10% raw (→ $535.-/t)	14.249	1.580	63.3	–4.863	**–16.49**
–10% raw (→ $438-/t)	23.975	0.939	106.5	+4.863	**+16.49**
+20% raw	9.387	2.399	41.7	–9.726	**–32.97**
–20% raw	28.838	0.781	128.1	+9.726	**+32.97**

Raw materials dominate VCOP; a ±10% swing in RM shifts NPV by roughly **±$16.5 MM**.

3.3 Utilities Cost Sensitivity (holds price, raw materials, FCOP, FCI fixed)

Scenario	OP(_y) (MM$)	PBP (y)	ROI (%)	ΔEBIT (MM$/y)	Approx ΔNPV (MM$)
Base	19.112	1.178	84.9	0.000	0.000
+10% utilities (→ $15.09/t)	18.975	1.187	84.3	–0.137	**–0.47**
–10% utilities (→ $12.35/t)	19.249	1.170	85.5	+0.137	**+0.47**
+20% utilities	18.838	1.195	83.6	–0.274	**–0.93**

Scenario	OP(_y) (MM$)	PBP (y)	ROI (%)	ΔEBIT (MM$/y)	Approx ΔNPV (MM$)
–20% utilities	19.387	1.162	86.1	+0.274	**+0.93**

Utilities are a small fraction of VCOP in this case; even ±20% utilities moves NPV by **< $1 MM**.

3.4 Capital Cost (FCI) Sensitivity (holds operations fixed)

Scenario	FCI (MM$)	OP(_y) (MM$)	PBP (y)	ROI (%)	Approx ΔNPV (MM$)
Base	22.520	19.112	1.178	84.9	0.000
+10% FCI	24.772	19.112	1.296	77.2	**–2.252**
–10% FCI	20.268	19.112	1.060	94.3	**+2.252**
+20% FCI	27.024	19.112	1.415	70.7	**–4.504**
–20% FCI	18.016	19.112	0.942	106.1	**+4.504**

NPV shifts **~one-for-one** with FCI (we conservatively ignore the modest change in depreciation tax shields). PBP/ROI are **sensitive** to FCI even when operations are unchanged.

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4) Interpretation & "What Matters Most"

- **Product price** has the **largest** impact: ±10% price moves NPV by **±$25 MM**, cuts PBP from **1.18 → 0.85 y** on the upside and stretches to **1.93 y** on the downside.
- **Raw-material cost** is the **second-largest lever**: ±10% RM shifts NPV by **±$16.5 MM** and materially changes PBP/ROI.
- **Capital cost** directly affects **PBP and ROI** (denominator effect) and moves NPV roughly one-for-one.
- **Utilities** are comparatively minor at this scale—price hedging here yields little value relative to RM or product marketing.

Ranking of drivers by NPV sensitivity (this plant):

1. **Product price** 2) **Raw materials** 3) **Capital cost (FCI)** 4) **Utilities**

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5) How to Reproduce (Formulas you can paste)

Operating profit (annual)

$$
\boxed{\text{OP}_y = \left(P - \text{VCOP}_{\$/t}\right) Q_y - \text{FCOP}_y}
$$

PBP

$$
\boxed{\text{PBP} = \dfrac{\text{FCI}}{\text{OP}_y}}
$$

ROI

$$
\boxed{\text{ROI} = \dfrac{\text{OP}_y}{\text{FCI}}\times 100\%}
$$

NPV change (approx.) for a change in price/costs

$$
\boxed{\Delta\text{NPV} \approx 0.65 \times \big[Q_y(\Delta P - \Delta \text{VCOP}_{\$/t})\big]\times 5.216}
$$

NPV change (approx.) for a change in FCI

$$
\boxed{\Delta\text{NPV} \approx -\Delta \text{FCI}}
$$

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6) Conclusion (what to say in your report)

- The project's economics are **most sensitive to market forces** (methanol price) and **feedstock strategy** (H_2/CO_2 costs).
- **Capital discipline** materially improves PBP/ROI but has a smaller effect on NPV than equivalent relative moves in price or raw-material costs.
- **Utilities** have a **second-order** effect.
- Managing price exposure (contracts, offtake, or futures) and securing favorable **H_2/CO_2** terms will **dominate** profitability relative to tweaking utility tariffs.

Self-Evaluation of the AI Workflow

AI used: ChatGPT (GPT-5 Thinking)

This project asked the AI to develop a complete economic analysis for a methanol plant—from total capital cost through CCOP, NPV, and discounted cash-flow rate of return—while retaining all prompts and outputs. Here I reflect on what the AI did well and where it struggled, using this assignment's deliverables as the frame of reference.

The AI's strongest contribution was its organization. It consistently produced report-ready sections with clear headings, definitions, and equations, then demonstrated each formula with worked examples. That structure made the analysis auditable: readers could see the variables, follow the algebra, and verify the arithmetic without deciphering hidden assumptions. In capital estimation, the AI moved smoothly from purchased equipment costs to an installed total using an explicit ISBL/OSBL/E&H/contingency split, and it cross-checked that outcome with a Lang-factor sanity check. On the operating side, it separated variable and fixed costs carefully, which is a common stumbling block for students; the fixed-cost framework relied on conventional percentages (maintenance, supervision, overhead, taxes and insurance) but kept those factors transparent and editable.

Cash-flow modeling was another area of strength. The AI set out a disciplined after-tax construction: EBIT before depreciation derived from revenues and costs; MACRS schedules applied with half-year convention; taxes calculated on taxable income; and annual cash flow recovered as net income plus depreciation. That approach prevented double-counting and made the later metrics—NPV at a stated discount rate and DCFROR—both traceable and repeatable. The sensitivity analysis was particularly helpful: rather than scattershot scenarios, the AI prioritized levers that matter most for this process (product price and raw-material costs) and quantified their effects on PBP, ROI, and NPV using compact, defensible calculations. These results gave practical guidance: manage methanol offtake and hydrogen/CO_2 sourcing first; utilities and minor capital tweaks matter less.

Despite these strengths, the analysis still depended heavily on assumptions. In the absence of vendor quotes, the AI relied on "typical" factors for installation, maintenance, overheads, and wages. Those are justifiable at screening level but can swing results by millions of dollars. Similarly, market prices for methanol, hydrogen, and carbon dioxide vary by region and contract structure. The AI offered reasonable benchmarks, yet without dated, location-specific citations or actual site contracts, the CCOP and margins remain provisional. Another source of ambiguity is vocabulary: terms like "gross profit," "ROI," and "CCOP" are not universal. The AI stated its conventions, but if an instructor expects depreciation inside COGS or ROI on total investment rather than fixed capital, reported values will shift. A brief up-front alignment with the course rubric would reduce that risk.

There were also practical issues of precision and scope. Small rounding differences accumulated across tables because the narrative was not tied to a single spreadsheet backbone. For professional reporting, one calculation source should feed every table to eliminate reconciliation noise. In scope, the AI kept the model unlevered and omitted working capital, salvage value, and financing structure. That choice is acceptable for an early screen, but it tends to overstate IRR and NPV relative to a fuller project finance treatment. If the assignment requires those items, the current analysis would need to be extended.

To mitigate these limitations, several steps are advisable. First, lock definitions at the outset: what counts as gross profit, how ROI is measured, which tax rate and project life to use, and whether to include working capital or salvage. Second, replicate the AI's equations in a spreadsheet or short Python script and link all tables to the same cells, ensuring consistent rounding and easy updates. Third, anchor price assumptions to dated sources or contracts and include a ±10–20% sensitivity around those anchors to communicate risk rather than false precision. Fourth, reconcile capital with Aspen Economic Evaluation on a common scope (purchased vs. installed) so that hand and software estimates are comparable.

In sum, the AI excelled at building a clean, teachable, and auditable framework, and it highlighted the dominant economic drivers with clarity. Its main weaknesses stem from assumption gravity, input sourcing, and definition alignment—not from the underlying cash-flow mechanics. With firmed-up inputs, a single computational backbone, and a brief rubric alignment, the AI's outputs are robust enough to support a credible feasibility-level decision and a strong academic submission.

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